



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
GB/T 16483-2008, GB/T 17519-2013

**Product Name** SDS Solution 20% (w/v)

**Revision date** 30-Sep-2024

**Revision Number** 1.1

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** SDS Solution 20% (w/v)

**Catalogue Number(s)** 1610418

### Other means of identification

**Pure substance/mixture** Mixture

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd.  
1st and 2nd Floor, Lumpini 1 Building  
239/2, Rajdamri Road, Lumpini,  
Pathumwan, Bangkok 10330  
Thailand

**Technical Service** +66 2 652 8313  
ctsthailand@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Hong Kong: 800-968-793

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

## SECTION 2: Hazards identification

### **Emergency Overview**

Irritating to skin  
Risk of serious damage to eyes

**Appearance** aqueous solution **Physical state** Liquid **Odour** Odourless

### Classification of the substance or mixture

|                                                       |            |
|-------------------------------------------------------|------------|
| Acute toxicity - Inhalation (Dusts/Mists)             | Category 4 |
| Skin corrosion/irritation                             | Category 2 |
| Serious eye damage/eye irritation                     | Category 1 |
| Specific target organ toxicity - Single exposure      | Category 3 |
| Hazardous to the Aquatic Environment - Acute Hazard   | Category 3 |
| Hazardous to the Aquatic Environment - Chronic Hazard | Category 3 |

**Label elements****Signal word**

Danger

**Hazard statements**

Harmful if inhaled  
Causes skin irritation  
Causes serious eye damage  
May cause respiratory irritation. May cause drowsiness or dizziness  
Harmful to aquatic life with long lasting effects

**Precautionary statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapours/spray  
Use only outdoors or in a well-ventilated area  
Wash face, hands and any exposed skin thoroughly after handling  
Avoid release to the environment  
Wear protective gloves/protective clothing/eye protection/face protection  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Call a POISON CENTRE or doctor if you feel unwell  
IF ON SKIN: Wash with plenty of water and soap  
If skin irritation occurs: Get medical advice/attention  
Take off all contaminated clothing and wash it before reuse  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTRE or doctor

**Storage**

Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Physical and chemical hazards**

Not applicable.

**Health hazards**

Immediate Health Effects: Harmful. Effects of exposure (inhalation, ingestion or skin contact) to substance may be delayed.  
Causes skin irritation (pain, redness and swelling). Risk of serious damage to eyes. Impairment of vision.  
Chronic effects: Not applicable.

**Environmental hazards**

This material is a water pollutant. Keep out of drains, sewers, ditches and waterways. Minimise use of water to prevent environmental contamination

**Other hazards which do not result in classification**

No information available

**SECTION 3: Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name         | Weight-% | CAS No.  |
|-----------------------|----------|----------|
| Sodium lauryl sulfate | 20 - 35  | 151-21-3 |

**SECTION 4: First aid measures****Description of necessary first aid measures**

|                                                                           |                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                                                     | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                |
| <b>Inhalation</b>                                                         | Remove to fresh air. Get medical attention immediately if symptoms occur. IF exposed or concerned: Get medical advice/attention. If symptoms persist, call a doctor. If breathing has stopped, give artificial respiration. Get medical attention immediately.                                                   |
| <b>Eye contact</b>                                                        | Get immediate medical advice/attention. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area.                                                 |
| <b>Skin contact</b>                                                       | Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.                                                                                                                                                                           |
| <b>Ingestion</b>                                                          | Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Get medical attention.                                                                                                                                               |
| <b><u>Most important symptoms and effects, both acute and delayed</u></b> | Burning sensation. Inhalation of high vapour concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Coughing and/ or wheezing. Difficulty in breathing.                                                                                                                     |
| <b><u>For emergency responders</u></b>                                    | Avoid contact with skin, eyes or clothing. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Avoid breathing vapours or mists. Use personal protective equipment as required. See section 8 for more information. |
| <b><u>Note to doctors</u></b>                                             | Treat symptomatically.                                                                                                                                                                                                                                                                                           |

**SECTION 5: Firefighting measures****Extinguishing media**

|                                                            |                                                                                                                                    |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                        | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                      | No information available.                                                                                                          |
| <b><u>Specific hazards arising from the chemical</u></b>   | No information available.                                                                                                          |
| <b><u>Special protective actions for fire-fighters</u></b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## SECTION 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                                                                    |                                                                                                                                                                                            |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b>                                        | Avoid contact with skin, eyes or clothing. Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Avoid breathing vapours or mists. |
| <b>Other information</b>                                           | Refer to protective measures listed in Sections 7 and 8.                                                                                                                                   |
| <b>For emergency responders</b>                                    | Use personal protection recommended in Section 8.                                                                                                                                          |
| <b><u>Environmental precautions</u></b>                            | Prevent further leakage or spillage if safe to do so.                                                                                                                                      |
| <b><u>Methods and material for containment and cleaning up</u></b> | Prevent further leakage or spillage if safe to do so. Pick up and transfer to properly labelled containers.                                                                                |
| <b><u>Precautions to prevent secondary hazards</u></b>             | Clean contaminated objects and areas thoroughly observing environmental regulations.                                                                                                       |

## SECTION 7: Handling and storage

|                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Precautions for safe handling</u></b>                                | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse. Ensure adequate ventilation. Avoid breathing vapours or mists. In case of insufficient ventilation, wear suitable respiratory equipment. Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. See Section 8 for information on appropriate personal protective equipment. |
| <b><u>Conditions for safe storage, including any incompatibilities</u></b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Incompatible materials</b>                                              | Strong acids. Strong bases. Strong oxidising agents.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## SECTION 8: Exposure controls/personal protection

### **Occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Note** See section 16 for terms and abbreviations

### **Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### **Monitoring and observation processes**

No applicable information was found.

### **Engineering controls**

Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

|                                |                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection            | Tight sealing safety goggles.                                                                                                                                            |
| Skin and body protection       | Wear suitable protective clothing. Long sleeved clothing.                                                                                                                |
| Hand protection                | Wear suitable gloves. Impervious gloves.                                                                                                                                 |
| Respiratory protection         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| General hygiene considerations | Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product.                             |

**SECTION 9: Physical and chemical properties****Information on basic physical and chemical properties**

|                 |                          |
|-----------------|--------------------------|
| Appearance      | aqueous solution         |
| Colour          | colourless               |
| Physical state  | Liquid                   |
| Odour           | Odourless                |
| Odour threshold | No information available |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      |                   | None known              |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | > 100 °C          |                         |
| Flash point                             | 170 °C            |                         |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapour pressure                         | No data available | None known              |
| Relative vapour density                 | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | 248 °C            |                         |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |
| <b>Additional information</b>           |                   |                         |
| Explosive properties                    | Not applicable    |                         |
| Oxidising properties                    | Not applicable    |                         |

**SECTION 10: Stability and reactivity**

|                                           |                                 |
|-------------------------------------------|---------------------------------|
| <u>Stability</u>                          | Stable under normal conditions. |
| <u>Possibility of hazardous reactions</u> | None under normal processing.   |

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

Conditions to avoid Excessive heat.Incompatible materials Strong acids. Strong bases. Strong oxidising agents.Hazardous decomposition products None known based on information supplied.**SECTION 11: Toxicological information**Acute toxicityNumerical measures of toxicity

No information available

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 6,440.00 mg/kg

ATEmix (inhalation-dust/mist) 4.875 mg/l

Unknown acute toxicity

0 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

**Component Information**

| Chemical name         | Oral LD50            | Dermal LD50            | Inhalation LC50                      |
|-----------------------|----------------------|------------------------|--------------------------------------|
| Sodium lauryl sulfate | = 1288 mg/kg ( Rat ) | = 200 mg/kg ( Rabbit ) | > 3900 mg/m <sup>3</sup> ( Rat ) 1 h |

**Skin corrosion/irritation** Classification based on data available for ingredients. Irritating to skin.**Serious eye damage/eye irritation** Classification based on data available for ingredients. Causes burns. Risk of serious damage to eyes.**Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.**Germ cell mutagenicity** Based on available data, the classification criteria are not met.**Carcinogenicity** Based on available data, the classification criteria are not met.**Reproductive toxicity** Based on available data, the classification criteria are not met.**Specific target organ toxicity — single exposure** May cause respiratory irritation, May cause drowsiness or dizziness.**Specific target organ toxicity — repeated exposure** Based on available data, the classification criteria are not met.**Aspiration hazard** Based on available data, the classification criteria are not met.**SECTION 12: Ecological information**Ecotoxicity Harmful to aquatic life with long lasting effects.

| Chemical name         | Algae/aquatic plants | Fish                      | Crustacea                    |
|-----------------------|----------------------|---------------------------|------------------------------|
| Sodium lauryl sulfate | EC50: =53mg/L (72h,  | LC50: 15 - 18.9mg/L (96h, | EC50: =1.8mg/L (48h, Daphnia |

|  |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|  | Desmodesmus subspicatus)<br>EC50: 30 - 100mg/L (96h,<br>Desmodesmus subspicatus)<br>EC50: =117mg/L (96h,<br>Pseudokirchneriella subcapitata)<br>EC50: 3.59 - 15.6mg/L (96h,<br>Pseudokirchneriella subcapitata) | Pimephales promelas)<br>LC50: 8 - 12.5mg/L (96h,<br>Pimephales promelas)<br>LC50: 22.1 - 22.8mg/L (96h,<br>Pimephales promelas)<br>LC50: 4.3 - 8.5mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =4.62mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =4.2mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =7.97mg/L (96h,<br>Brachydanio rerio)<br>LC50: 9.9 - 20.1mg/L (96h,<br>Brachydanio rerio)<br>LC50: 4.06 - 5.75mg/L (96h,<br>Lepomis macrochirus)<br>LC50: 4.2 - 4.8mg/L (96h,<br>Lepomis macrochirus)<br>LC50: =4.5mg/L (96h, Lepomis<br>macrochirus)<br>LC50: 5.8 - 7.5mg/L (96h,<br>Pimephales promelas)<br>LC50: 10.2 - 22.5mg/L (96h,<br>Pimephales promelas)<br>LC50: 6.2 - 9.6mg/L (96h,<br>Pimephales promelas)<br>LC50: 13.5 - 18.3mg/L (96h,<br>Poecilia reticulata)<br>LC50: 10.8 - 16.6mg/L (96h,<br>Poecilia reticulata)<br>LC50: =1.31mg/L (96h, Cyprinus<br>carpio) | magna) |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|

**Persistence and degradability** No information available.

**Bioaccumulative potential** There is no data for this product.

#### Component Information

| Chemical name         | Partition coefficient |
|-----------------------|-----------------------|
| Sodium lauryl sulfate | 1.6                   |

**Mobility in soil** No information available.

### SECTION 13: Disposal considerations

**Waste chemicals** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### SECTION 14: Transport information

**IMDG** Not regulated  
**Transport in bulk according to** No information available  
**Annex II of MARPOL and the IBC**  
**Code**

IATA Not regulated

JT/T 617 Not regulated

**Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

**SECTION 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations****Law of the People's Republic of China on Prevention and Control of Occupational Diseases**

Catalogue of occupational hazard factors:

Not applicable.

Catalogue of occupational diseases:

Not applicable.

**Regulations on the Control over Safety of Hazardous Chemicals**

Inventory of hazardous chemicals

Not applicable.

GB 18218-2009 Identification of major hazard installations for dangerous chemicals

Not applicable

**List of hazardous chemicals under priority management**

Not applicable

**Regulations on Labour Protection in Workplaces Where Toxic Substances Are Used**

Inventory of highly toxic goods

Not applicable

**Regulations for Environmental Management on the First Import of Chemicals and the Import and Export of Toxic Chemicals**

List of toxic chemicals severely restricted for import and export in China

Not applicable

**Measures for the Environmental Management of New Chemical Substances**

**IECSC - China Inventory of Existing Chemical Substances** Contact supplier for inventory compliance status.

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**SECTION 16: Other information**

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety

**Revision date** 30-Sep-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Abbreviations and acronyms**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

TWA TWA (time-weighted average)

STEL

STEL (Short Term Exposure Limit)

Ceiling Maximum limit value

Sk\*

Skin designation



C Carcinogen

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Disclaimer**

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**End of Safety Data Sheet**